Feedim

VERSION 1.2

INSTRUCTIONS FOR USE



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1. DESCRIPTION AND SPECIFICATIONS

Intended use

The Feedim system is a combination of a hardware device, the Feedim device, and server software that connects enteral nutrition pumps to third-party software.

The Feedim device reads data from an enteral nutrition pump (without controlling or altering the functions or parameters of the connected pump) and transmits it via Wi-Fi directly to the nutrition management software Nutrow or indirectly through the Electronic Health Record (EHR) or Electronic Medical Record (EMR).

Transmitted data are also visible in the Feedim software where, using the information from the pump (the actual delivered volume) and the delivered product characteristics, you can easily know, check, and monitor the quantity of nutrients that your patient receives over time.

The Feedim system is designed for professional use only.

Please note that this manual focuses on Feedim device and server software only. For more information about how the data provided by Feedim are integrated in Nutrow, please refer to the Nutrow Instructions for use manual.

Description

For hospitals and homecare companies that want to monitor enteral nutrition pumps, Feedim automatically provides the necessary data for clinical nutrition follow-up and helps save data entry time and prevent data entry errors for healthcare professionals.



Manufacturer information

- Feedim 1.2 GTIN: 05404025300315
- Manufacturer location:

  Dim3
  Rue de Louveigné, 188/d
  4052 Beaufays
  Belgium

- Website: www.dim3.com

Environmental specifications

Ambient operating temperature	+10° to +40°C
Ambient operating humidity	35% to 85% relative humidity (RH), non-condensing
Ambient oxygen concentration	Less than 25% at one atmosphere or partial pressure up to 27,5 kPa for higher atmospheric pressures
Storage and transportation temperature	-20°C to +40°C
Cleaning and disinfection	Clean and disinfect the Feedim device and its accessories following use to prevent patient cross infection. To determine proper disinfection procedure in a particular institution, follow your local institution policy. Recommended use of medical cleaning and disinfecting solutions such as methylated spirits or isopropyl alcohol with a soft cloth for a few seconds. Please note that some cleaning and disinfecting agents based on formaldehyde, potassium or sodium hydroxide, ammonia and ammonium chloride are not recommended as they may damage or discolour the Feedim housing which may void the warranty in this regard. A comprehensive list with the resistance per chemicals is available on request.

Hardware specifications

Dimensions	62 (L) x 42 (W) x 93.5 (H) mm
Input voltage range	90-264 Vac
Input frequency	50-60 Hz
Weight	114 g
Case material	PC-ABS plastic
Wi-Fi	802.11 b/g/n (802.11n up to 150 Mbps)
Bluetooth	Bluetooth v4.2 BR/EDR and BLE specification
Battery	Lithium polymer 3.7 V 980 mAh
Expected battery lifetime	After 300 complete cycles of charge, the battery will still have 80% of its nominal capacity
Connector	RJ12
IP Protection	IP 22
Devices per box	6
Box dimensions	357 (L) x 235 (W) x 135 (H) mm
Box weight	Containing 6 devices: • with RJ45 cables: 1925 g • with DB9 cables: 2000 g
Box component part list	Each box contains: • 6 Feedim devices • 6 AC/DC adapters XP POWER ACM06US05 • 6 input plugs for the region/country of usage • 6 cables RJ45-RJ12 or DB9-RJ12 depending of the pump type
Lot number	The lot number is printed on the box label
Type of medical device	Medical Device Class I

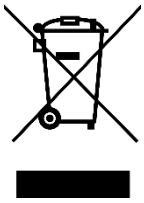
Please note that the Feedim device may require up to 24 hours to warm or to cool from its storage conditions until it is ready to use.

Please note that the Feedim device must be used only with the XP POWER ACM06US05 AC/DC adapter provided in the box.

Software specifications

Interface	Web application supporting Chrome, Firefox, Safari and Edge
Languages	Multi-language support
Setup	Server hosted on premises or in a cloud environment (SaaS)
Supported databases	Support of standard databases (Oracle, MySQL, PostgreSQL, SQL Server)
Privacy	Privacy by design and anonymized user data
Cybersecurity	State-of-the-art cybersecurity standards and best practices that fit within a healthcare organization / Defence in Depth (DiD) strategy: • PKI and authentication token security infrastructure • OAuth 2.0 token-based authentication and authorization open standard • HTTPS with TLS 1.2/1.3 security protocol Please refer to the “Technical Installation Guide” for a full security description.

Disposal



The applicable standards for electrical waste and electronic devices state that they must be recycled.

Note that the battery is not removable and cannot be replaced by the user.

If you are no longer using this equipment or if it cannot be repaired, do not throw it away in your usual household garbage can. Return it to the vendor or contact the customer service department in charge of the device for further information.

Instructions for use

To obtain a paper copy of this manual, please contact Dim3 (info@dim3.com or +32 4 252 35 00). The paper copy will be sent within 7 days.

Symbols

	Manufacturer
	Catalogue number
	Separate collection for electrical and electronic equipment waste
	Caution, risk of electric shock
	Date of manufacture
	Protected against solid foreign objects 12.5 mm (0.5 in) in diameter or more; protected against vertical water drops when enclosure tilted up to 15°
	Follow operating instructions
	Caution
	Serial number
	Batch code
	Medical device

2. SAFETY INFORMATION

READ THE ENTIRE MANUAL BEFORE INSTALLING AND OPERATING THE FEEDIM DEVICE

Installation

This guide assumes that the **Feedim Web Application** has been installed and configured properly by the manufacturer, their distributor, or their representative.

For documentation related to installation and IT administration, please refer to the Technical Installation Guide (available upon request).

Contraindications

- Do not use the Feedim device in areas where there is a risk of explosions (e.g., in the presence of flammable anaesthetics) to avoid fire and spontaneous combustion.
- Do not use the Feedim device in areas with strong magnetic fields (e.g., Magnetic Resonance Imaging environments).
- The Feedim device only supports a limited number of pump models (see list below). Do not use the Feedim device with a different pump model.
    - REF “FE0001-6-xx-DB” contains connectivity cables with RJ12 connector (on the Feedim side) and DB9 connector compatible only with the following pump models:
        - Nestlé Compat Ella®
        - Nutricia Flocare Infinity®
        - Cardinal Health Kangaroo ePump® (formerly Covidien/Medtronic)
    - REF “FE0001-6-xx-RJ” contains connectivity cables with RJ12 connector (on the Feedim side) and RJ45 connector compatible only with the following pump model:
        - Fresenius Kabi Applix Smart® enteral feeding pump.

Warnings

- Do not use the Feedim device for any purpose other than what is described in this manual.
- Risk of injuries or malfunction: Do not open or modify the Feedim device.
- Risk of injuries or malfunction: Do not insert anything into the Feedim device (except for the connection cables provided with the Feedim device and pumps).
- Risk of injuries or malfunction: Check the Feedim device and cable before use. Do not use the Feedim device if it has been damaged or altered. Do not use the Feedim device if the rubber presents excessive signs of wear and/or if the clamp’s spring becomes too weak, preventing Feedim to be fixed firmly on the IV-pole.

Cautions

- Risk of injury: Read the entire user manual before using the Feedim device.
- Risk of injury: Use only the Feedim AC Power Adapter provided with the Feedim device. Using an alternate consumer-style AC power adapter may damage the charging circuit and the battery of the Feedim device. Do not attempt to attach the country-specific AC adapter to AC power without attaching it to the AC adapter body.
- Risk of injury: Do not submerge the Feedim device nor its cables or sterilize them in an autoclave to avoid failure or injury.
- Risk of injury: Do not put the Feedim device and its cables in the dishwasher to avoid failure or injury.
- Risk of injury: Do not clean the Feedim device while it is plugged in to avoid electrical shock.
- Risk of injury: Strangulation resulting from baby or child entanglement in cables.
- Risk of injury: Twinge resulting from inserting fingers or other parts of a human body between the clamp and the rear side of the Feedim device.
- Risk of injury: Rely only on data reported by a Feedim device if its status is “GREEN” (see sections “Monitoring” and “Patients List”). Healthcare professionals may be misled if the information provided is not up to date and may have impact on patient’s health.
- Risk of injury: The Feedim device must be used in a vertical position and attached to the IV pole with its clamp (the IV pole diameter should be 18–36 mm) to prevent it from falling.
- Loss of functionality: Always check that the Feedim software has been properly integrated with the healthcare organization information system before use.
- Loss of functionality: Always check that the pairing has been carried out properly before using the Feedim Web Application (see section “Pairing”).
- Loss of functionality: Do not use the Feedim device if the Feedim Server and cybersecurity infrastructure is not yet in place to avoid security breaches.
- Loss of functionality: The Feedim battery capacity is only an estimate. If you are not sure that enough capacity remains for your intended use, connect the Feedim device to AC power.
- Loss of functionality: If the Feedim device has been stored for a long period of time, it should be plugged in before use.
- Loss of functionality: The extensive use of harsh cleaners, disinfectants, or isopropyl alcohol may damage the Feedim housing. Follow the instructions in the Environmental Specifications chapter.

Serious incident

If any serious incident which may be attributed to Feedim occurs, it should be reported to Dim3 (vigilance@dim3.com) and to the AFMPS (Agence Fédérale des Médicaments et des Produits de Santé in Belgium).

3. OVERVIEW

Initialization

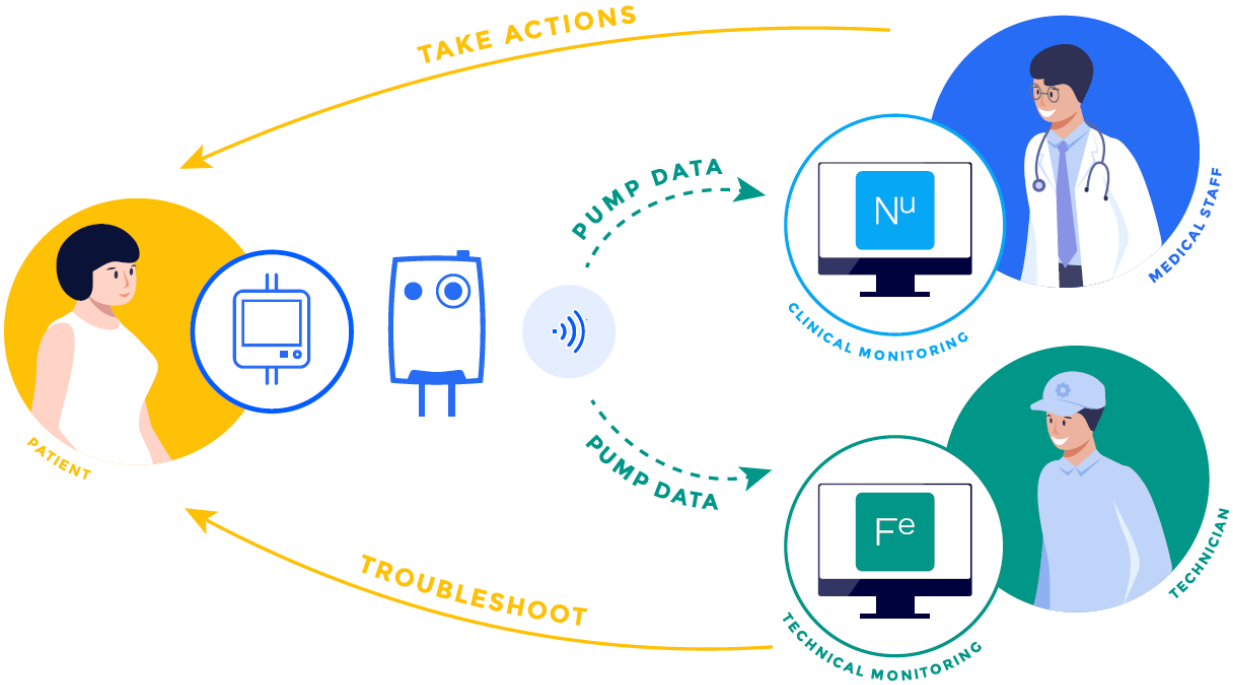
Before using the Feedim device and the **Feedim Web Application**, you must follow these installation steps: integration, configuration, installation, and pairing.



Note: In this manual, we will describe how to carry out the bedside installation and pairing of the Feedim device. For more information about the IT integration and configuration of the Feedim system, please refer respectively to the Technical Installation Guide and to the Configuration Guide.

Normal use

Feedim is a combination of hardware device and server software. Each Feedim device captures the data from the pump to which it is connected via a cable and transmits it wirelessly to the server. The server captures the data from all the devices and makes it available to healthcare professionals and technicians in the **Feedim Web Application** and in **Nutrow**.



4. INSTALLATION

The system IT integration and configuration must be performed before the Feedim device can be installed at bedside.

4.1 Installing the Feedim device

1. Install and connect the power supply.



Note: Your Feedim is equipped with a backup battery. This battery has a capacity of about one month when fully charged and operating with the standard configuration settings (a wake-up and connection every 15 minutes). Nevertheless, we highly recommend that you keep your Feedim powered at all times.

2. Connect your pump to the Feedim using the cables described in the “Connecting the Feedim device” section.

3. Clamp your Feedim onto the pump support next to the pump as illustrated.



4.2 Connecting the Feedim device to the pump

Connect the Dim3 connectivity cable to your Feedim (RJ12 connector as illustrated) and to your pump or pump cable.



According to your pump model, you received one of the following cables:

- Dim3 connectivity cable DB9-RJ12 (FE2010-DB)
- Dim3 connectivity cable RJ45-RJ12 (FE2010-RJ)

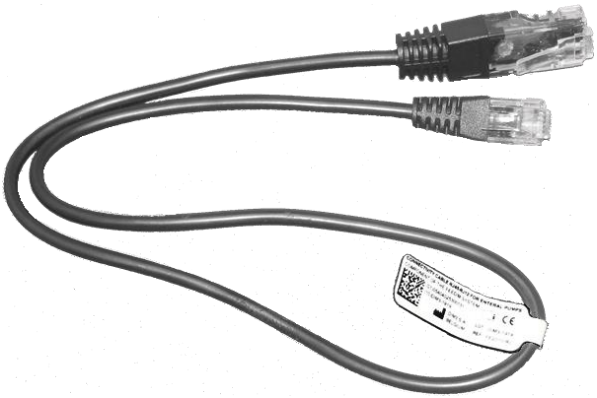
Cables

Pump model	**Dim3 cable**	**Pump manufacturer cable**
Fresenius Applix Smart®	RJ45-RJ12 (FE2010-RJ)	—
Nutricia Infinity Flocare®	DB9-RJ12 (FE2010-DB)	Nutricia Infinity Flocare® data cable
Cardinal Health Kangaroo ePump®	DB9-RJ12 (FE2010-DB)	Kangaroo ePump® communication cable
Nestlé Compat Ella®	DB9-RJ12 (FE2010-DB)	Nestlé Compat Ella® PDMS cable

Note: When connecting Feedim device to the pump and/or to AC power, check if it is well connected and make sure that you let enough free space to easily disconnect or unplug the device if needed.

Fresenius Applix Smart® pump

Feedim cable: Dim3 connectivity cable RJ45-RJ12 (FE2010-RJ)





Pump cable: no cable needed, direct connection to the pump

Nutricia Infinity Flocare® pump

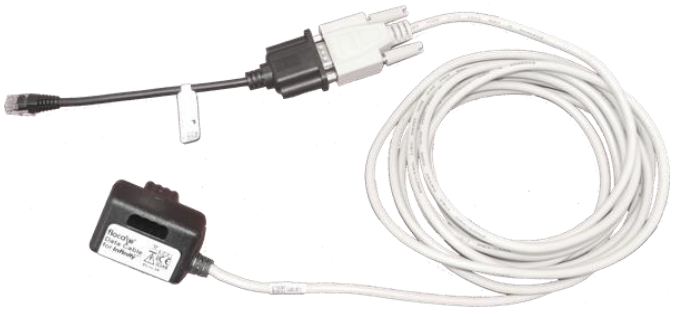


Feedim cable: Di ty cable DB9-RJ12 (FE2010-DB)





Pump cable: Nutricia Infinity Flocare® data cable

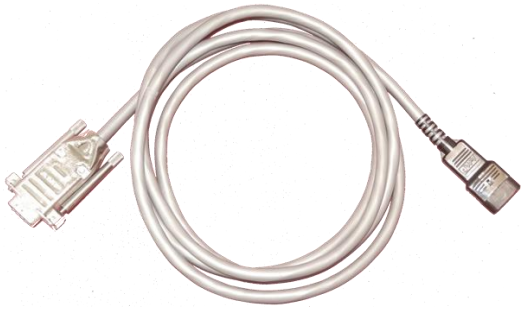


Cardinal Health Kangaroo ePump® pump

Feedim cable: Dim3 connectivity cable DB9-RJ12 (FE2010-DB)



Pump cable: Kangaroo ePump® communication cable



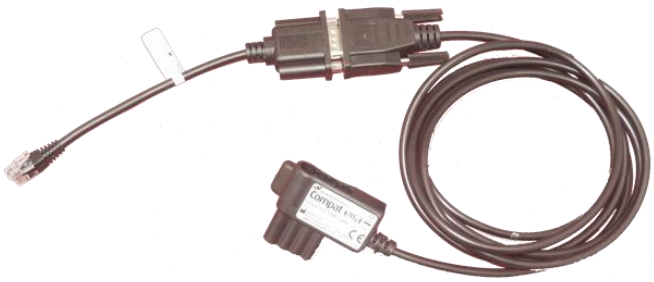


Nestlé Compat Ella® pump

Feedim cable: Dim3 connectivity cable DB9-RJ12 (FE2010-DB)



Pump cable: Nestlé Compat Ella® PDMS cable





5. PAIRING

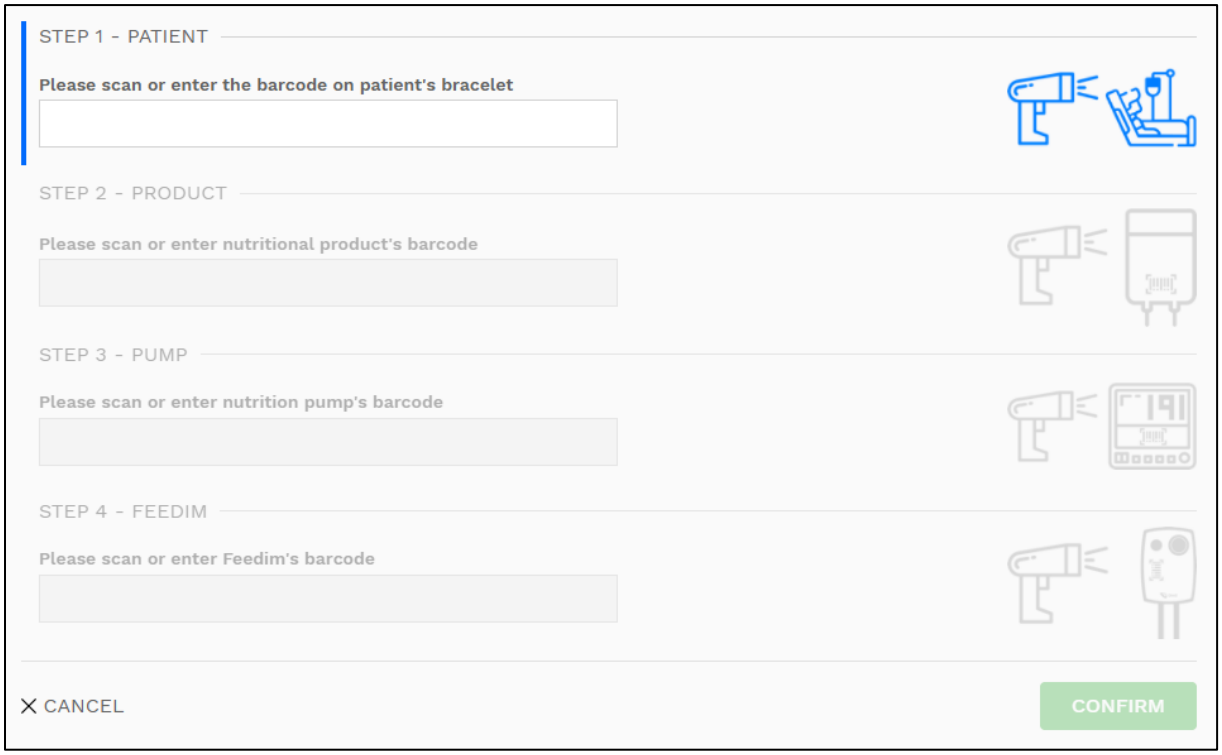
To open this tab in the Feedim Web Application, you must have an Admin or Nurse profile.

The pairing process refers to the operation of scanning the patient identification bracelet (or other), the feeding pump, nutrition products and the Feedim to ensure the traceability of all elements. This process is mandatory before using the Feedim and must be processed every time there is a change of one of the elements (patient, feeding pump, nutrition products and Feedim).

To perform the pairing, you need a configured Feedim device (for Configuration, please refer to the Feedim Technical Installation Guide) and a computer with access to the **Feedim Web Application**. To facilitate this process, we also recommend that you use a barcode scanner.

Warning: The pump must be reset before performing the pairing process.

1. On the computer, run the **Feedim Web Application** (and log in).
2. Open the **Feedim pairing wizard**.
3. Scan or enter the patient identifier.
4. Scan or enter the product identifier.
5. Scan or enter the pump identifier.
6. Scan or enter the Feedim identifier.
7. Validate the pairing.

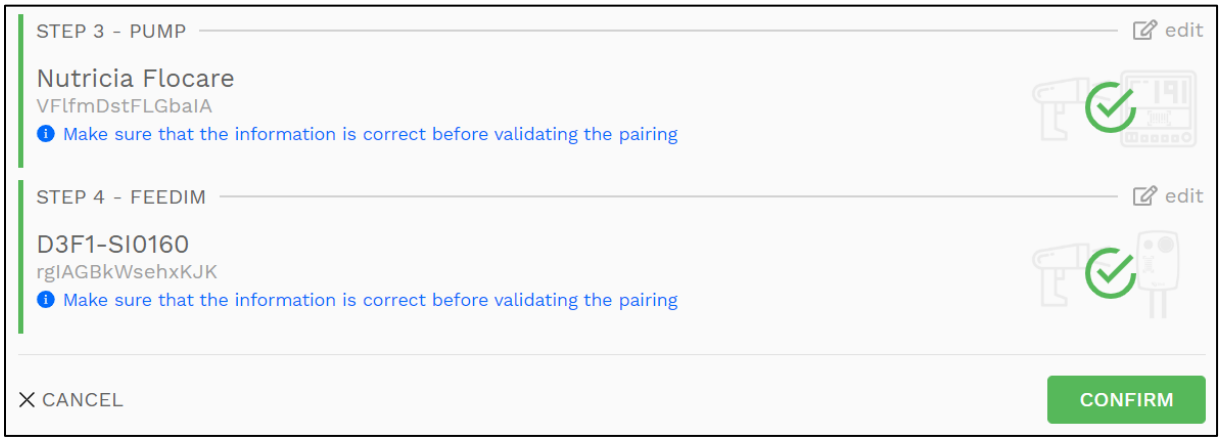


To modify an entry, click the **Edit** button.

To stop the pairing process, click the **Cancel** button.

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At configuration, the healthcare organisation can choose to authorise the simplified pairing process: if a pairing has already been performed for a given patient-product combination, the pump identifier and Feedim identifier fields will be automatically filled in if you scan the same patient and product identifiers during the pairing process.

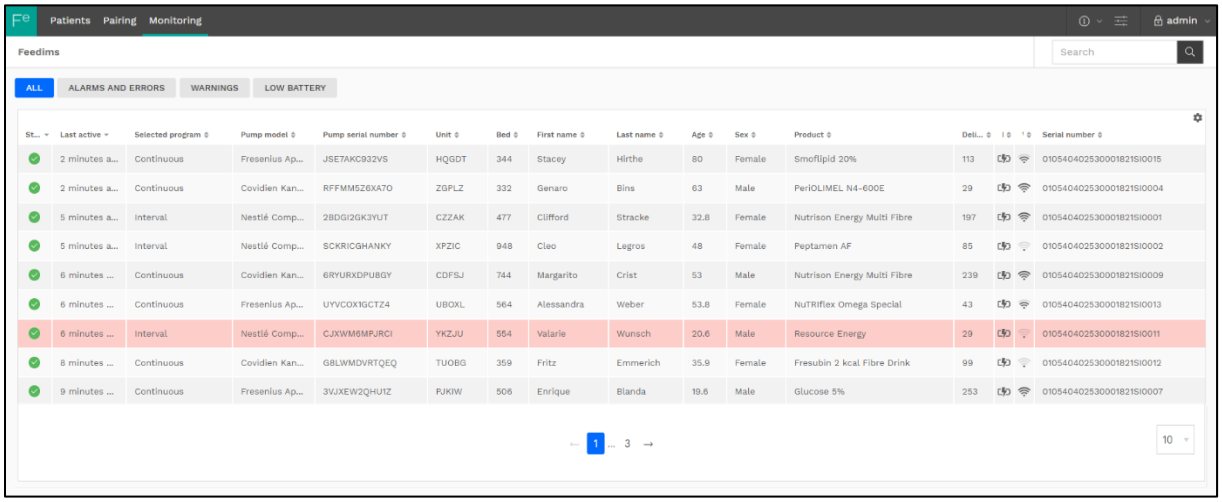


6. MONITORING

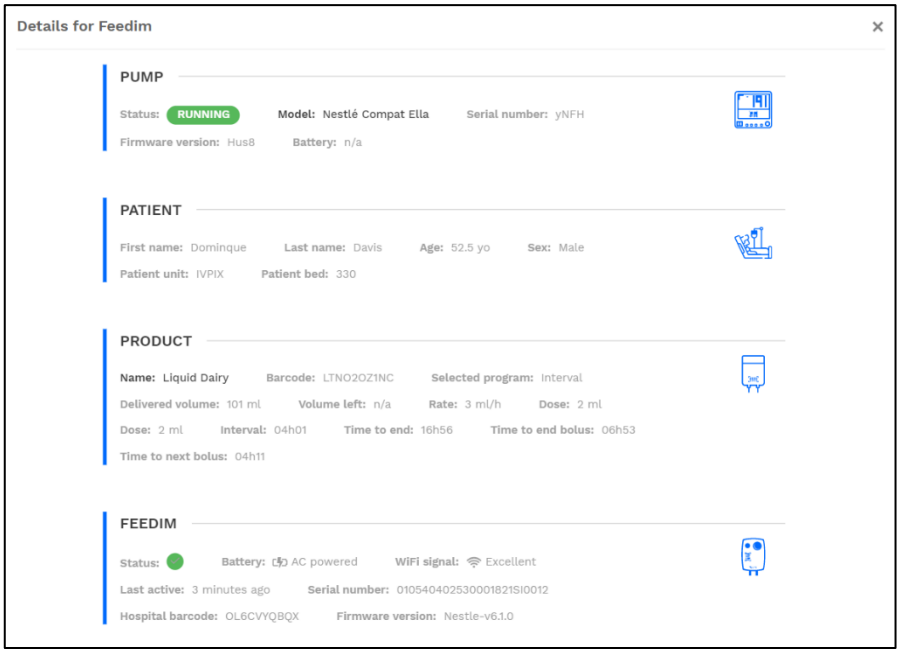
To open this tab in the Feedim Web Application, you must have an Admin or Nurse or Technician profile.

Use the **Monitoring** tab of the **Feedim Web Application** to check and monitor all your Feedim devices.

1. On the computer, run the **Feedim Web Application** (and log in).
2. Open the **Monitoring** tab or click the Feedim icon.



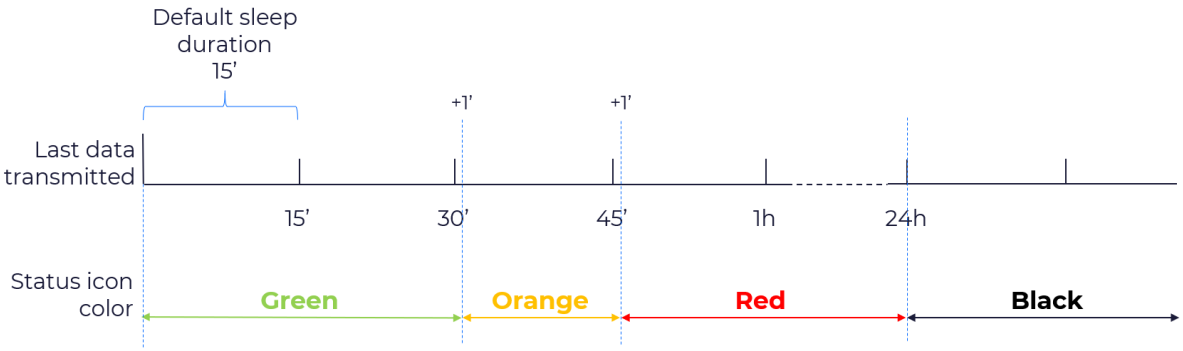
3. Click on a Feedim to open the following window displaying its current state and system environment.



The Feedim status icon colour on the **Monitoring** screen is based on the elapsed time since the last piece of data was received. As such, it is linked to the sleep duration of the Feedim device. It indicates the last time that the Feedim device was active and sent information to the **Feedim Web Application,** and is managed by the Admin.

Feedim status icon Colour	**The last piece of data was transmitted…**
Green	… less than **[sleep duration × 2] + 1 minute** ago
Orange	… between **[sleep duration × 2] + 1 minute** and **[sleep duration × 3] + 1 minute** ago
Red	… between **[sleep duration × 3] + 1 minute** and **24 hours** ago
Black	… more than **24 hours** ago
Grey	The Feedim device was never paired or never sent information

Refer to the following timeline for a precise interpretation (example with default Feedim device sleep duration of 15 mins.):



Note: None of the status icon colour requires a prompt or immediate response.

Orange, red, or black status does not necessarily mean that the Feedim device is malfunctioning. A Wi-Fi network outage could also be the reason for such a status (please refer to the section "Troubleshooting" for more information).

The **Monitoring** tab also indicates the status of the Feedim battery:

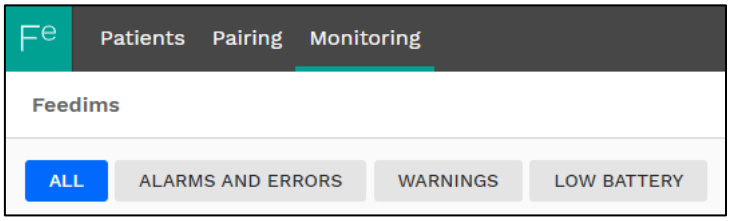
Icon	Battery status
	AC powered
	Fully charged
	Three-quarters charged
	Half charged
	Quarter charged
	Empty battery

Please note that the first time you will access the **Monitoring** tab, all the columns will be displayed by default.

To select the columns you want to be displayed, click on the **Cog Wheel** button on the top right of the list and check / uncheck the columns depending on your needs.

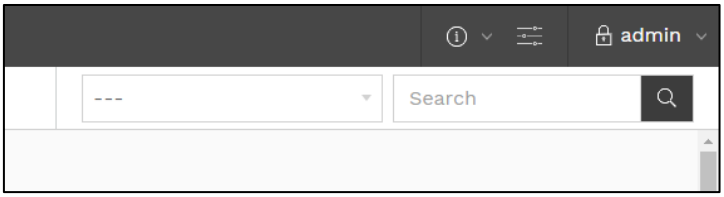
Filters

To filter the pumps with alarms and errors or with warnings, select the corresponding filter on the top of the list. This filter is also available to display the Feedims with low battery (Feedim with battery status icon of "Empty battery").



Please note that if a filter (alarms and errors, warnings, low battery, or area) and/or a sort had been chosen, these filter/sort will be applied by default. Filters can be cumulated (if an area is selected and then the filter "Warnings", the monitoring screen will only display the Feedims that have warnings in the chosen area).

To filter the Feedims by area, select the chosen area in the dropdown list on the top right of the monitoring screen.



To reset a filter, click on "All" or "---" on the area dropdown.

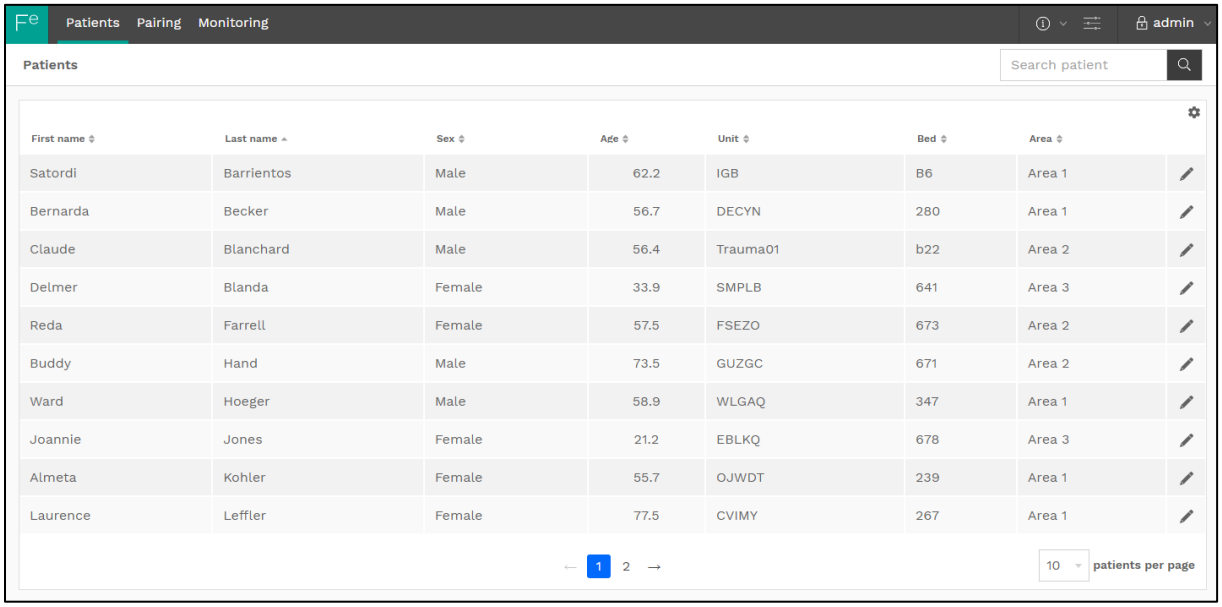
Please note that depending on the institution configuration, this feature may not be available.

7. PATIENTS LIST

To open this tab in the Feedim Web Application, you must have an Admin or Nurse profile.

Use the **Patients List** tab of the **Feedim Web Application** to check and monitor all your patients.

1. On the computer, run the **Feedim Web Application** (and log in).
2. Open the **Patients List** tab.



To edit the patient's barcode (used for the pairing process), click on the Edit button. Please note that depending on the institution configuration, the edition of the barcode could be not available.

To select the columns you want to display, click on the **Cog Wheel** button on the top right of the list and check / uncheck the columns depending on your needs

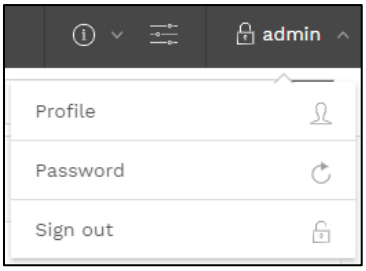
Please note that the first time you will access the **Patients List**, all the columns will be displayed by default.

Please note that depending on the institution configuration, the area column could not be available.

8. ADMINISTRATION

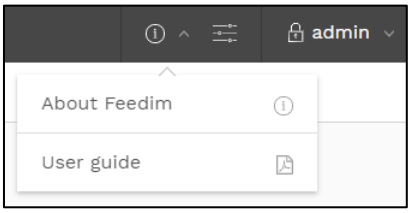
Use the **Administration** screen of the **Feedim Web Application** to manage your profile.

1. On the computer, run the **Feedim Web Application** (and log in).
2. Click the button with the name of the current user to open the **Administration** window.
    - In the **Profile** section, change your interface language.
    - In the **Password** section, change your login password.
    - Click **Sign out** to log out of the application.



Use the Information button to get more information about Feedim.

- Click the **Info** then **About Feedim** icon to open the **About** window.
- Click the **Info** then **User guide** to open the **Feedim user guide.**



9. FEEDIM DEVICE LED COLOURS AND MEANINGS

- At the automatic wake-up (when the Feedim is active and sends information to the **Feedim Web Application)**, if everything is okay: the Feedim device LED turns **green** for 1 second.
- At automatic wake-up, if something goes wrong: the Feedim device LED turns **green** for 1 second, and then display the error code after its cycle (10 to 45 seconds) by blinking **light blue** for one second one, two, three or four times (see below).

Numbers of light blue blinks	Meanings
One time	The Feedim device failed to connect to the Wi-Fi
Two times	The Feedim device failed to communicate with the Feedim Server
Three times	The Feedim device failed to communicate with the pump
Four times	The Feedim device has encountered another error

Note: None of the LED colour requires a prompt or immediate response.

Note: To wake up the Feedim device manually and display the last error code, press the button for 1 second.

10. TROUBLESHOOTING

Condition	Cause	Correction
Feedim device		
The Feedim device LED doesn't blink	The battery may be discharged, and the device may be not powered	Wake Feedim device up by pressing the button for 1 second Connect the power supply to the Feedim device and turn it on
The Feedim device LED blinks light blue for one second one time	The Feedim device failed to connect to Wi-Fi	Verify that the Wi-Fi network is up and running Check the Feedim device is within the Wi-Fi range Contact your technical support
The Feedim device LED blinks light blue for one second two times	The Feedim device failed to connect to the Feedim Server or the pairing failed	Perform a new pairing Contact your technical support
The Feedim device LED blinks light blue for one second three times	The Feedim device failed to communicate with the pump	Check that the pump is compatible with Feedim Check the cable connection between Feedim and the pump Check that the pump is on / powered / plugged to the AC/DC power Perform a new pairing
The Feedim device LED blinks light blue for one second four times	The Feedim device has encountered another error	Contact your technical support
The Feedim device LED blinks light blue every 100ms	The Feedim device is downloading its configuration from the Feedim Server	Wait up to 3 minutes to let the Feedim device download its configuration Contact your technical support
Can't connect AC/DC power to the Feedim device	Cable termination is not suitable for Feedim device input plug	Ensure it's the right AC/DC power adapter Ensure it's the right adapter
Can't connect pump connectivity cable	Cable termination is not suitable for Feedim communication port	Ensure not confounding the two plugs Ensure you're using the right cable (refer to section Installation of these IFUs)

Condition	Cause	Correction
		Ensure you're not trying to connect the pump to the Feedim device with the manufacturer data cable directly (w/out the Dim3 cable) – except for the Fresenius Applix Smart
Feedim device doesn't stick to IV pole	IV pole is too big or too small	Check your IV pole is not too big (36mm diameter max) or too small (18mm min) Contact your technical support
Monitoring		
Feedim status icon remains black or red or grey	Last data was received too long ago / never received	Ensure pairing is done properly Check if there is an error code on the Feedim device led
User can't login to the Feedim Web Application	Wrong username / Wrong password / User profile not allowed to access the Monitoring	Check with Feedim System Admin
Information is not updated	Feedim cannot retrieve its data from the nutrition pump	Check if the pump is powered Check if the pump is plugged in Check if the connectivity cable is plugged Ensure pairing is done properly Check if there is an error code on the Feedim device LED
The error message "The Feedim cannot connect to the pump. Please check that you paired the Feedim to the pump it's physically connected. Error code: 50" is displayed	The Feedim cannot connect to the pump	Check that you paired the Feedim device to the pump it's physically connected
The error message "There is an error with the pump communication. Please check that the pump is powered and that the cable is correctly connected between the Feedim and the pump. Error code: 119" is displayed	There is an error with the pump communication	Check that the pump is powered and that the cable is correctly connected between the Feedim device and the pump
Patient list		

Condition	Cause	Correction
User can't login	Wrong username / Wrong password / User profile not allowed to access the Patient list	Check with Feedim System Admin
Pairing wizard		
User can't login	Wrong username / Wrong password / User profile not allowed to access the pairing wizard	Check with Feedim System Admin
Scanned patient / product / pump / Feedim doesn't exist	The patient / product / pump / Feedim has not been configured	Check with Feedim System Admin
Can't access / see the barcode on nutrition pump	The barcode is hidden or missing	Check with Feedim System Admin to add a new barcode on the pump

11. WARRANTY

General conditions of the warranty

Dim3 warrants the Feedim device against any defect in parts or workmanship that renders it unfit for the intended use for a period of 2 years starting on the date of the delivery.

This warranty only covers the repair and replacement of defective parts. It does not cover any consequential damages that may result from such defects in parts or workmanship of the Feedim device.

The action in warranty must be brought by the purchaser within 15 days from the discovery of the defect.

Limitations of the warranty

The warranty does not apply in the following circumstances:

- The Feedim device has not been used in accordance with the instructions described in the user manual.
- The Feedim device has deterioration caused by improper transport, storage, or improper handling.
- The Feedim device has undergone adaptations or repairs carried out by unqualified persons.
- The battery of the Feedim device has been replaced by a different battery from the one specifically chosen by Dim3 (note that the battery is not removable by the user).
- The defective Feedim device is a "second hand" device, resulting from a resale between professionals, with the exception of resales carried out in the context of mergers / acquisitions of companies.

In these cases, Dim3 will prepare a repair estimate.

Battery and accessories

This warranty does not cover battery elements, charger (and accessories). These items are subject to separate warranty conditions. For more information on these specific warranties, please contact Dim3.